

Semantic Technologies

Assignment 2 (Group 10)

a) How many triples / statements are in the ontology?

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT (COUNT(*) AS ?tripleCount)
WHERE {
  ?s ?p ?o .
}
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query is pasted into the editor and has been executed. The results section shows a single row with the value 1,577,796,732 for the variable ?tripleCount.

SPARQL.dblp.org beta

Getting Started Resources

```
1 PREFIX dblp: <https://dblp.org/rdf/schema#>
2 PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
3
4 PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
5 SELECT (COUNT(*) AS ?tripleCount)
6 WHERE {
7   ?s ?p ?o .
8 }
9
```

Execute Download Share Format Clear cache Analysis Examples 3. Context sensitive suggestions

Automatically add names to result

Query results: 1 lines found 0.0ms in total 0.0ms for computation 0.0ms for resolving and sending

	?tripleCount
1	1,577,796,732

Screenshot recorded on: 21 June 2026.

b) How many publications are in the ontology?

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT (COUNT(?pub) AS ?publicationCount)
WHERE {
  ?pub rdf:type dblp:Publication .
}
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query is entered in the editor and is as follows:

```
1 PREFIX dblp: <https://dblp.org/rdf/schema#>
2 PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
3
4 PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
5 SELECT (COUNT(?pub) AS ?publicationCount)
6 WHERE {
7   ?pub rdf:type dblp:Publication .
8 }
```

Below the query editor, there are buttons for 'Execute', 'Download', 'Share', 'Format', 'Clear cache', 'Analysis', and 'Examples'. A dropdown menu shows '3. Context sensitive suggestions' and a checkbox for 'Automatically add names to result' is unchecked.

The 'Query results:' section shows the following statistics: 1 lines found, 14ms in total, 13ms for computation, and 1ms for resolving and sending. The results table has one column, '?publicationCount', and one row with the value 8,597,634.

	?publicationCount
1	8,597,634

Screenshot recorded on: 21 June 2026.

c) What is the name of the author with the Orcid-ID 0000-0002-5515-7158?

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>

SELECT ?authorName
WHERE {
  ?author dblp:orcid ?orcidURI ;
          dblp:primaryCreatorName ?authorName .
  FILTER(CONTAINS(STR(?orcidURI), "0000-0002-5515-7158"))
}
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query is entered in the main text area and is as follows:

```
1 PREFIX dblp: <https://dblp.org/rdf/schema#>
2
3 SELECT ?authorName
4 WHERE {
5   ?author dblp:orcid ?orcidURI ;
6           dblp:primaryCreatorName ?authorName .
7   FILTER(CONTAINS(STR(?orcidURI), "0000-0002-5515-7158"))
8 }
```

Below the query editor, there are buttons for 'Execute', 'Download', 'Share', 'Format', 'Clear cache', 'Analysis', and 'Examples'. A dropdown menu shows '3. Context sensitive suggestions'.

The 'Query results:' section shows the following table:

	?authorName
1	Ralph Bergmann

At the top of the results section, there are statistics: '1 lines found', '119ms in total', '118ms for computation', and '1ms for resolving and sending'.

Screenshot recorded on: 21 June 2026.

d) How many authors have a Wikipedia page?

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT (COUNT(DISTINCT ?author) AS ?wikipediaAuthorCount)
WHERE {
  ?author rdf:type dblp:Creator ;
          dblp:wikipedia ?wikiPage .
}
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query is entered in the main text area and is highlighted. Below the query area are buttons for 'Execute', 'Download', 'Share', 'Format', 'Clear cache', 'Analysis', and 'Examples'. A checkbox for 'Automatically add names to result' is also present. The 'Query results' section shows a table with one column, '?wikipediaAuthorCount', and one row with the value '9,337'. The table is titled '1 lines found' and shows timing information: '11ms in total', '11ms for computation', and '0.0ms for resolving and sending'.

	?wikipediaAuthorCount
1	9,337

Screenshot recorded on: 21 June 2026.

e) Which papers contain “CBR” in their title (limit to 10)?

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?paper ?title
WHERE {
  ?paper rdf:type dblp:Publication ;
        dblp:title ?title .
  FILTER(CONTAINS(STR(?title), "CBR"))
}
LIMIT 10
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query results are displayed in a table with two columns: ?paper and ?title. The results are limited to 10 rows. The interface includes a top navigation bar with 'Getting Started' and 'Resources' links. Below the query input area, there are buttons for 'Execute', 'Download', 'Share', 'Format', 'Clear cache', 'Analysis', and 'Examples'. A checkbox for 'Automatically add names to result' is also present. The query results table shows the following data:

	?paper	?title
1	TreDCG08	Flexible Querying Techniques Based on CBR.
2	ArcosML11	Using Introspective Reasoning to Improve CBR Sys[...]
3	CheethamCG01	Applications of Soft CBR at General Electric.
4	Diaz-AgudoG07	An Ontological Approach to Develop Knowledge Int[...]
5	PietroGPS19	J2CBROKER as a Service: A Service Broker Simulat[...]
6	KrishnanMSK18	Experiment: Investigating Feasibility of Coexist[...]
7	MalandraSACH22	Community CBRS Networks - What You Need to Know.
8	BahlsR07	Explanation Support for the Case-Based Reasoning[...]
9	Hammond92	Reasoning as Remembering: The Theory and Practic[...]
10	OuWLC05	Interactive Knowledge Validation and Query Refin[...]

Screenshot recorded on: 21 June 2026.

f) Co-authors of Ralph Bergmann and Alexander Schultheis

SPARQL query

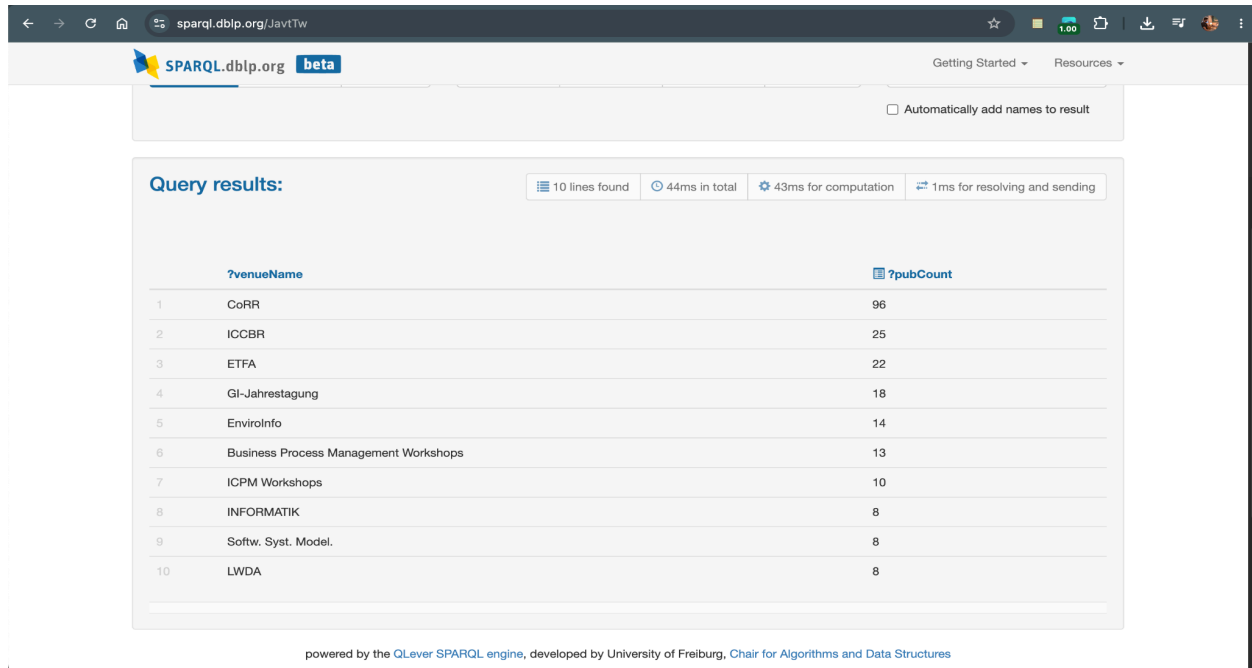
```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?venueName (COUNT(DISTINCT ?otherPub) AS ?pubCount)
WHERE {
  # 1. Find shared papers between Bergmann and Schultheis
  ?sharedPub dblp:authoredBy ?b1, ?b2, ?coAuthor .
  ?b1 dblp:primaryCreatorName "Ralph Bergmann" .
  ?b2 dblp:primaryCreatorName "Alexander Schultheis" .

  # 2. Exclude the target authors from being counted as co-authors
  FILTER(?coAuthor != ?b1 && ?coAuthor != ?b2)

  # 3. Find other publications by these co-authors and their venues
  ?otherPub dblp:authoredBy ?coAuthor ;
            dblp:publishedIn ?venueName .
}
GROUP BY ?venueName
ORDER BY DESC(?pubCount)
LIMIT 10
```

Result Screenshot:



The screenshot shows the SPARQL.dblp.org interface. The query results are displayed in a table with two columns: ?venueName and ?pubCount. The table lists 10 venues, sorted by publication count in descending order. The venues and their counts are: CoRR (96), ICCBR (25), ETFA (22), GI-Jahrestagung (18), EnviroInfo (14), Business Process Management Workshops (13), ICPM Workshops (10), INFORMATIK (8), Softw. Syst. Model. (8), and LWDA (8).

	?venueName	?pubCount
1	CoRR	96
2	ICCBR	25
3	ETFA	22
4	GI-Jahrestagung	18
5	EnviroInfo	14
6	Business Process Management Workshops	13
7	ICPM Workshops	10
8	INFORMATIK	8
9	Softw. Syst. Model.	8
10	LWDA	8

powered by the QLever SPARQL engine, developed by University of Freiburg, Chair for Algorithms and Data Structures

Screenshot recorded on: 21 June 2026.

g) How many publications are in the ontology?

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT (COUNT(DISTINCT ?paper) AS ?dfkiTrierPubs)
WHERE {
  ?paper rdf:type dblp:Publication ;
         dblp:authoredBy ?author ;
         dblp:yearOfPublication ?year .
  ?author dblp:primaryCreatorName ?name .
  # Ensure the author matches either professor exactly
  FILTER(?name = "Ralph Bergmann" || ?name = "Ingo J. Timm")
  # Use alphanumeric string evaluation to check if the year is 2020 or greater
  FILTER(STR(?year) >= "2020")
}
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query is pasted into the editor and has been executed. The results section shows a single result for the variable `?dfkiTrierPubs` with the value 127. The interface includes a toolbar with buttons for Execute, Download, Share, Format, Clear cache, Analysis, and Examples. The footer indicates the query is powered by the QLever SPARQL engine, developed by the University of Freiburg, Chair for Algorithms and Data Structures.

SPARQL.dblp.org beta

Getting Started Resources

```
6 WHERE {
7   ?paper rdf:type dblp:Publication ;
8         dblp:authoredBy ?author ;
9         dblp:yearOfPublication ?year .
10
11   ?author dblp:primaryCreatorName ?name .
12
13   # Ensure the author matches either professor exactly
14   FILTER(?name = "Ralph Bergmann" || ?name = "Ingo J. Timm")
15
16   # Use alphanumeric string evaluation to check if the year is 2020 or greater
17   FILTER(STR(?year) >= "2020")
18 }
```

Execute Download Share Format Clear cache Analysis Examples 3. Context sensitive suggestions

Automatically add names to result

Query results: 1 lines found 27,644ms in total 27,643ms for computation 1ms for resolving and sending

	?dfkiTrierPubs
1	127

powered by the QLever SPARQL engine, developed by University of Freiburg, Chair for Algorithms and Data Structures

Screenshot recorded on: 21 June 2026.

h) German researchers with ≥ 800 publications

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?author ?name ?affiliation (COUNT(DISTINCT ?paper) AS ?pubCount)
WHERE {
  ?author rdf:type dblp:Creator ;
          dblp:primaryCreatorName ?name ;
          dblp:primaryAffiliation ?affiliation .
  ?paper dblp:authoredBy ?author .
  FILTER(CONTAINS(STR(?affiliation), "Germany"))
}
GROUP BY ?author ?name ?affiliation
HAVING (?pubCount >= 800)
ORDER BY DESC(?pubCount)
LIMIT 10
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org web interface. The query results are displayed in a table with the following columns: ?author, ?name, ?affiliation, and ?pubCount. The results are sorted by the number of publications in descending order. The table shows 10 results, with the first row being Nassir Navab with 1,784 publications. The interface also includes a status bar at the bottom indicating the query was powered by the QLever SPARQL engine.

	?author	?name	?affiliation	?pubCount
1	NassirNavab	Nassir Navab	Technical University Munich, Germany	1,784
2	2265	Robert Schober	Friedrich-Alexander University Erlangen-Nürnberg[...]	1,433
3	AloisKnoll	Alois C. Knoll	Technical University Munich, Department of Compu[...]	1,257
4	WilIMPvanderAalst	Wil M. P. van der Aalst	RWTH Aachen University, Chair of Process and Dat[...]	1,252
5	7133	Andreas K. Maier	University of Erlangen-Nuremberg, Department of [...]	1,223
6	2478	Daniel Rueckert	Technical University of Munich, Germany	1,169
7	2492	Giuseppe Caire	Technical University of Berlin, Department of Te[...]	1,132
8	GFettweis	Gerhard P. Fettweis	Dresden University of Technology, Germany	1,088
9	119	Bernhard Schölkopf	Max Planck Institute for Intelligent Systems, Tü[...]	1,069
10	HermannNey	Hermann Ney	RWTH Aachen University, Germany	950

powered by the QLever SPARQL engine, developed by University of Freiburg, Chair for Algorithms and Data Structures

Screenshot recorded on: 21 June 2026.

i) Ralph Bergmann's co-authors ranked by first-authored publications

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?coAuthorName ?affiliation ?orcidStr (COUNT(DISTINCT ?firstAuthoredPaper) AS
?firstAuthorCount)
WHERE {

    // 1. Find Ralph Bergmann's direct co-authors
    ?sharedPaper dblp:authoredBy ?rb , ?coAuthor .
    ?rb dblp:primaryCreatorName "Ralph Bergmann" .
    FILTER(?coAuthor != ?rb)
    ?coAuthor dblp:primaryCreatorName ?coAuthorName .

    // 2. Match papers where they are the FIRST author via the signature path
    ?firstAuthoredPaper dblp:hasSignature ?sig .
    ?sig dblp:signatureCreator ?coAuthor ;
        dblp:signatureOrdinal 1 .

    // 3. Pull optional metadata fields safely
    OPTIONAL { ?coAuthor dblp:primaryAffiliation ?affiliation . }
    OPTIONAL { ?coAuthor dblp:orcid ?orcidStr . }
}
GROUP BY ?coAuthorName ?affiliation ?orcidStr
ORDER BY DESC(?firstAuthorCount)
LIMIT 12
```

Result Screenshot:

sparql.dblp.org/2nTLbR

SPARQL.dblp.org beta

Getting StartedResources

Query results:

12 lines found7ms in total6ms for computation1ms for resolving and sending

	?coAuthorName	?affiliation	?orcidStr	?firstAuthorCount
1	Yolanda Gil	University of Southern California, Information S[...]	0000-0001-8465-8341	89
2	Barry Smyth	University College Dublin, Insight Centre for Da[...]	0000-0003-0962-3362	87
3	Ulrich Furbach	University of Koblenz and Landau, Germany	0000-0002-3158-4446	74
4	Klaus Schmid	University of Hildesheim, Department of Computer[...]	0000-0002-4147-3942	69
5	Stefanie Rinderle-Ma	Technical University of Munich, Germany	0000-0001-5656-6108	56
6	Karl Branting	-	0000-0002-9362-495X	53
7	Manuela M. Veloso	Carnegie Mellon University, Pittsburgh, USA	0000-0001-6738-238X	53
8	Mirjam Minor	Goethe University of Frankfurt, Germany	0000-0002-6592-631X	52
9	Manfred Reichert	University of Ulm, Germany	0000-0003-2536-4153	49
10	Erica Melis	German Research Institute for Artificial Intelli[...]	-	47
11	Barbara Weber	University of St. Gallen, Switzerland	0000-0002-6004-4860	46
12	Klaus-Dieter Althoff	University of Hildesheim, Germany	0000-0002-7330-6540	45

Screenshot recorded on: 21 June 2026.

j) Publications of Ralph Bergmann where he is the sole author

SPARQL query

```
PREFIX dblp: <https://dblp.org/rdf/schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?title ?firstAuthorName ?numAuthors ?year
WHERE {
  // Link the paper to Ralph Bergmann and extract basic metadata
  ?paper dblp:authoredBy <https://dblp.org/pid/b/RalphBergmann> ;
    dblp:title ?title ;
    dblp:numberOfCreators ?numAuthors ;
    dblp:yearOfPublication ?year ;
    dblp:hasSignature ?sig .

  // Filter strictly for the first author's signature block
  ?sig dblp:signatureOrdinal 1 ;
    dblp:signatureCreator ?firstAuthor .

  // Retrieve the first author's actual name string
  ?firstAuthor dblp:primaryCreatorName ?firstAuthorName .
}
ORDER BY DESC(xsd:integer(STR(?year)))
LIMIT 1
```

Result Screenshot:

The screenshot shows the SPARQL.dblp.org interface. The query is pasted into the editor, and the 'Execute' button is highlighted. Below the editor, the query results are displayed in a table format. The table has four columns: ?title, ?firstAuthorName, ?numAuthors, and ?year. The results show one entry: 'Numbers Don't Lie: Hybrid Extraction and Validat[...]' by 'Mirko Lenz' with 4 authors, published in 2026.

?title	?firstAuthorName	?numAuthors	?year
Numbers Don't Lie: Hybrid Extraction and Validat[...]	Mirko Lenz	4	2026

Screenshot recorded on: 21 June 2026.